

Esperimento 6 - augmented dataset

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Extended Evaluation of the Attack-Day Prediction Model

Objective

Following the initial results presented in the previous experiment, the attack-day prediction model was further evaluated to determine whether its performance could be improved by increasing the diversity of the training dataset. The objective was to assess whether different attack scenarios would allow the model to learn more general patterns associated with vulnerable trading days.

Extended Dataset Construction

The dataset generation process remained identical to the previous experiment, with the exception that additional attack scenarios were included.

In particular, two new configurations were introduced:

- attacks evaluated using a **10-day evaluation window** instead of the original 20-day window;
- attacks targeting **AMZN** in addition to the previously considered GOOGL and AAPL assets.

The motivation behind these additions was to expose the model to a broader range of market conditions and attack outcomes, potentially allowing it to learn more informative relationships between observable market features and attack success.

Results

Despite the larger and more diverse dataset, the overall predictive performance of the model did not improve significantly. Similar to the previous experiment, the model was still unable to reliably identify vulnerable trading days using only market-derived information.

To further assess its practical usefulness, the trained model was integrated into the attack framework and used to select attack days on a previously unseen test period. Its performance was then compared against the timing policies introduced in the previous experiments, including random selection and heuristics based on market volatility, trend strength, and other observable characteristics.

The comparison revealed an interesting outcome. The ML-based timing policy proved to be relatively effective at identifying days on which the ATS was more susceptible to signal manipulation. Approximately **29%** of the attacks successfully forced the trading system to change its decision, making the learned policy the **second-best** timing strategy in terms of signal manipulation success.

However, successfully changing the trading signal did not necessarily translate into a meaningful degradation of trading performance. Only about **14%** of the successful attacks resulted in a reduction of portfolio performance over the subsequent **10–20 day** evaluation period, placing the ML-based policy **second from last** among the evaluated timing strategies.

Discussion

These results reinforce the conclusions drawn from the previous experiment. Although the model appears capable of identifying situations in which the ATS decision boundary is relatively fragile, these situations are not necessarily those in which manipulating the trading signal leads to economically significant consequences.

This suggests that **decision vulnerability** and **attack impact on portfolio performance** are related but distinct problems. Observable market information may contain signals indicating when an ATS is easier to manipulate, but these signals alone appear insufficient to identify the attacks that ultimately produce the greatest financial impact.

Overall, the extended experiments provide further evidence that predicting truly impactful attack opportunities from publicly observable market data remains a challenging task.